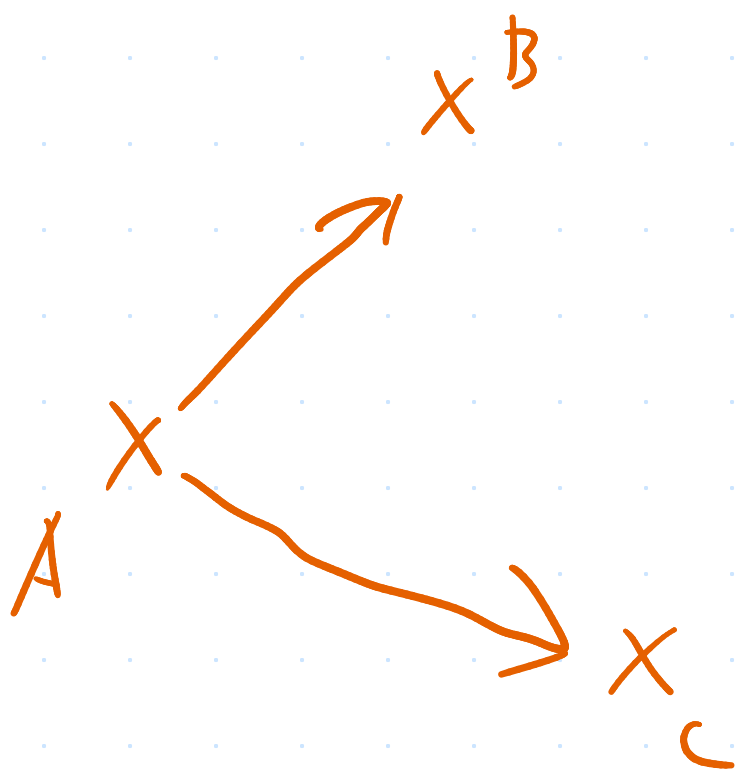


Parametrická rovnice

$$X = A + \vec{u}t + \vec{v}s$$

$$\vec{u} = \vec{AB}$$

$$\vec{v} = \vec{AC}$$



$$x = a_1 + u_1t + v_1s$$

$$y = a_2 + u_2t + v_2s$$

$$z = a_3 + u_3t + v_3s$$

$$A = [-1; 2; 1]$$

$$B = [0; 3; -2]$$

$$C = [5; 1; 3]$$

$$\vec{u} = \vec{AB} = (1; 1; -3)$$

$$\vec{v} = (6; -1; 2)$$

$$\rho: x = -1 + t + 6s$$

$$y = 2 + t - s$$

$$z = 1 - 3t + 2s$$

$$A = [2; -5; 6]$$

$$B = [1; 3; -2]$$

$$C = [3; 2; -1]$$

$$\vec{u} = (-1; 8; -8)$$

$$\vec{v} = (1; 7; -7)$$

$$\rho: x = 2 - t + s$$

$$y = -5 + 8t + 7s$$

$$z = 6 - 8t - 7s$$

$$x = a_1 + u_1t + v_1s$$

$$y = a_2 + u_2t + v_2s$$

$$z = a_3 + u_3t + v_3s$$

$$\vec{u} = \vec{AB}$$

$$\vec{v} = \vec{AC}$$

Zjistěte, zda bod K leží v rovině ρ

$$K = [-2; 3; 1]$$

$$\rho: \begin{aligned} x &= 2 - t + s \\ y &= -5 + 8t + 7s \\ z &= 6 - 8t - 7s \end{aligned}$$

$$1) -2 = 2 - t + s \Rightarrow -s - 2 = 2 - t$$

$$2) 3 = -5 + 8t + 7s \quad -s = 4 - t$$

$$3) 1 = 6 - 8t - 7s \quad s = t - 4$$

$$2) 3 = -5 + 8t + 7(t - 4)$$

$$3) 1 = 6 - 8t - 7(t - 4)$$

$$2) 3 = -5 + 8t + 7t - 28$$

$$3) 1 = 6 - 8t - 7t + 28$$

$$\left. \begin{aligned} 2) 12 &= 5t \Rightarrow t = \frac{12}{5} \\ 3) 11 &= 5t \Rightarrow t = \frac{11}{5} \end{aligned} \right\} \frac{12}{5} \neq \frac{11}{5} = K \notin \rho$$

$$\rho: \begin{aligned} x &= -t + 4s \\ y &= 2 + t - 3s \\ z &= 1 + t + 2s \end{aligned}$$

$$1) 2 = -t + 4s \Rightarrow t = 4s - 2$$

$$2) 1 = 2 + t - 3s$$

$$3) 5 = 1 + t + 2s$$

$$2) 1 = 2 + 4s - 2 - 3s$$

$$3) 5 = 1 + 4s - 2 + 2s$$

$$\left. \begin{aligned} 2) 1 &= s \\ 3) 6 &= 6s \end{aligned} \right\} 1 = 1 \Rightarrow P \in \rho$$

$$A = [0; 2; 1]$$

$$B = [-1; 3; 2]$$

$$C = [4; -1; 3]$$

$$P = [2; 1; 5]$$

$$\vec{u} = (-1; 1; 1)$$

$$\vec{v} = (4; -3; 2)$$

$$P \in \rho (?)$$